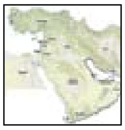


WiMax gaining ground in MENA nations
WiMax is gaining considerable ground in the Middle East and North Africa (MENA) nations, according to WiFi trackers. This is mostly because of a lack of penetration of fixed line internet services. The connections however, are mostly for business purposes

Still wait for WiMax
BSNL is still evaluating the tenders for WiMax services in India

be used for bandwidths up to 70Mbps by a large number of people simultaneously.

Orthogonal: This means mutually perpendicular. Here it applies to signals, that the signals are perpendicular to each other so that when one is at its peak, all others are zero.

Frequency Division: This is a means of transmitting multiple signals via a single medium, by using carriers of different frequencies. DSL is an example of how voice signals and internet connectivity signals are both carried in the same line.

Multiple access: This bit is obvious, a single access internet connection would be amazing surely, but the waiting in line for access would be quite annoying! Networking technologies have to ensure a feasible and stable method for allowing multiple people at access the service at the same time.

SOFDMA builds upon OFDMA which builds upon FDMA. SOFDMA is thus a specialised form of frequency division multiple access (FDMA). However, it is incompatible with the earlier technologies in some ways, and will require WiMAX 'broadcasters' to upgrade their equipments to support SOFDMA.

So, why SOFDMA, what does it do, and why is it important, that it was introduced despite compatibility issues.

Bandwidth, frequencies and data transfer

To transmit any kind of data it needs to first be modulated so that it can be carried over some kind of medium. Wired methods transmit data by sending electric signals of varying frequencies through the wire, which are modulated with the information to be sent. This modulation process involves the conversion of data which is composed of binary digits into a kind of electrical waveform.

In the modulation process, a base waveform or a carrier wave, is taken and modified with a pattern which is representative of the data being transferred. What quality of the carrier wave is modified depends on the modulation used. How much information a wave can carry depends greatly on the frequency of the waveform.

Different modulation schemes approach modulation in different ways in order to gain better and better efficiency. It is better to divide the same data and send it in parallel at multiple lower frequency bands than to transmit it over a single very high frequency band. The reason for this is simple, a single channel carrying data at 1 Mbps means that each binary digit will occur for just 1 micro-second! On the other hand, sending the same over 100 channels at 10 kbps would give us a much better duration per bit leading to easier detection.

The reason FDMA is superseded by OFDMA and OFDMA by SOFDMA is precisely because each improving technology brings with it better bandwidth efficiency, greater reach and better stability. The main concept used in all technologies remains frequency division, so let's have a look at FDMA from which they evolved.

FDMA

FDM or frequency division multiplexing is a way of transmitting multiple streams of data along the same medium (wired or wireless) by using different frequency bands for each channel. By keeping the frequency bands far enough apart, we can ensure that each channel can be filtered out separately. We can compare this to a room filled with people speaking in different languages. Although you may be able to hear everything, you will only be able to understand the languages you know.

In reality, we can look at the example of an FM Radio, where different channels come at different frequencies. One signal may be allotted a central frequency of 98.1 MHz with a bandwidth of 200 kHz, meaning that it will broadcast at frequencies between 98.0 MHz and 98.2 MHz, while another may be allocated a 200 kHz bandwidth around the 98.7 MHz. As such, each frequency band is separated by sufficient margins so that they don't overlap or interfere. On the receiving end, we can tune in the receiver to a particular frequency and filtering out the other signals, and boosting power at the selected frequency.

FDMA (Frequency Division Multiple Access), works by giving different users their own frequency bands, thus each user can communicate using a channel of their own.

This makes FDM quite inefficient, as a lot of bandwidth is wasted in the gaps used to separate different frequency bands. OFDM or Orthogonal Frequency Division Multiplexing packs in much more information by eliminating the requirement for these gap-bands.

OFDM and OFDMA

With OFDM (Orthogonal Frequency Division Multiplexing), sub-carriers in the signal wave can be overlapping without causing any interference. This is accomplished by making the sub-carriers in an OFDM exactly orthogonal to each other, meaning that while one is at its peak the others are all zero. The efficiency is greatly increased by this, as more data streams can now be transmitted using the same bandwidth.

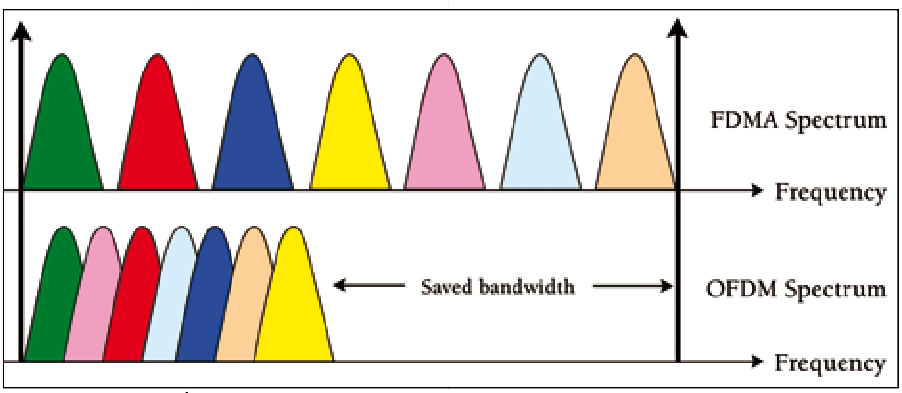
OFDM spreads out the data among multiple carriers, each of which is modulated separately. As there is now larger number of carriers to modulate, the effective system throughput is increased. Also since a large no of frequencies is used, some of the issues surrounding wireless communication such as multi-path, signal cancellation, and spectral interference are reduced to a great extent.

However in OFDM each channel is assigned to a single user. To enable multiple user access it is additionally used with Time Division Multiple Access (TDMA) or Frequency Division Multiple Access (FDMA), however each approach has its caveats.

OFDMA (Orthogonal Frequency Division Multiple Access) as its name suggests is designed for multiple user access. Unlike OFDM, OFDMA is capable of supporting multiple users on the same channel, by

FACT BOX

IEEE establishes and maintains the 802.16 standard which The WiMAX Forum uses to certify devices. IEEE does not test for standards compliance; as such WiMAX is not a standard in itself but rather a use of the IEEE Standard 802.16.



OFDM vs DMA comparison